

2-WEEK DEEP LEARNING PROJECT

# Summer School Project

Segmentation, Attribute Detection,  
and Anchored Findings Reports

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# Overview



- Medical Imaging
- Deep Learning
- Clinical AI

This two-week project builds a dermoscopy pipeline using the International Skin Imaging Collaboration (ISIC) dataset. You will develop two core computer vision components: (1) **lesion segmentation** to identify the lesion region, and (2) **lesion attribute detection** to localize five clinically meaningful dermoscopic patterns (pigment network, negative network, streaks, milia-like cysts, and globules). Using these outputs, you will complete a third task that focuses on controlled natural language generation: producing an **anchored findings report** in a strict JSON format plus a short, **human-readable summary**. An optional bonus track introduces **CLIP-based semantic retrieval** to find similar cases and use them to stabilize attribute predictions and improve report structure, while keeping all reported facts grounded in the model outputs.

# Why dermoscopy and why these tasks?



## Real medical images

Dermoscopy provides rich visual patterns with clinical relevance, offering authentic learning signals for medical AI applications.



## Pixel-level supervision

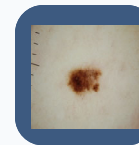
Mask annotations enable clear, interpretable learning objectives with direct visual feedback on model performance.



## Grounded outputs

Goal is auditable, evidence-based reports rather than free-form text, ensuring clinical trust and transparency.

### Project Workflow



Image



Findings



Report

# What you will build



## Task 1: Lesion Segmentation

Input dermoscopy image → Binary lesion mask



## Task 2: Attribute Detection

Input dermoscopy image → Five attribute masks (pigment network, negative network, streaks, milia-like cysts, globules)



## Task 3: Anchored Findings Report

Task 1 + Task 2 outputs → Structured JSON + Short report text



**Key Requirement:** Report must include the five medical terms (pigment network, negative network, streaks, milia-like cysts, globules) and be consistent with predicted evidence.

# Task 1: Lesion segmentation



## Input

Dermoscopy image  
(RGB)



## Output

Binary lesion mask (foreground/background)



## Evaluation

Dice coefficient and IoU (Intersection over Union)



**Note:** Accurate segmentation also supports downstream evidence feature extraction for the final report.

## Segmentation Example



(a) Input Image



(b) Ground Truth



(c) Prediction



(d) Error Map

## Task 2: Lesion attribute detection



### Output

Five attribute masks, one per dermoscopic term



### Purpose

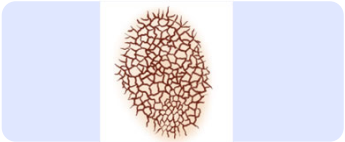
These terms serve as medical anchors for the report



### Evaluation

Per-attribute Dice + mean across all five

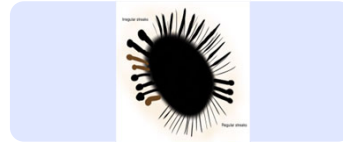
### Five Dermoscopic Attributes



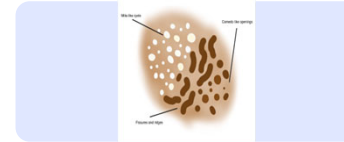
Pigment Network



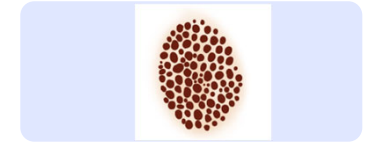
Negative Network



Streaks



Milia-like Cysts



Globules



**Challenge:** Some attributes are sparse and harder to detect



**Best practice:** Report per-attribute performance separately

# Task 3: Anchored dermoscopic findings report

## JSON Schema Output

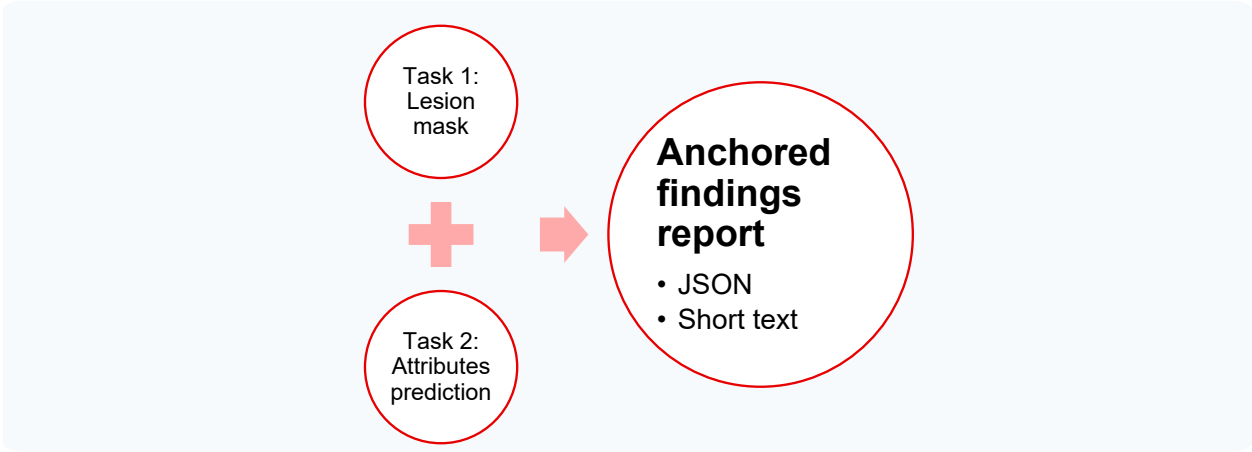
```
{
  "image_id": "0001234",
  "split": "val",
  "model_version": "task2_best_model",
  "attributes_order": [
    "pigment_network",
    "negative_network",
    "streaks",
    "milia_like_cysts",
    "globules"
  ],
  "outputs": {
    "presence": {
      "pigment_network": {
        "prob": 0.81,
        "status": "present"
      },
      "negative_network": {
        "prob": 0.12,
        "status": "absent"
      },
      "streaks": {
        "prob": 0.51,
        "status": "uncertain"
      },
      "milia_like_cysts": {
        "prob": 0.08,
        "status": "absent"
      },
      "globules": {
        "prob": 0.72,
        "status": "present"
      }
    }
  }
}
```

## Generated Report Text

The lesion is **moderate** with **irregular** borders. **Pigment network** is present; **negative network** is absent; **streaks** are uncertain; **milia-like cysts** are absent; **globules** are present.

## Consistency Checklist

- ✓ All five required terms present
- ✓ Statuses match JSON fields
- ✓ Evidence aligns with predicted masks



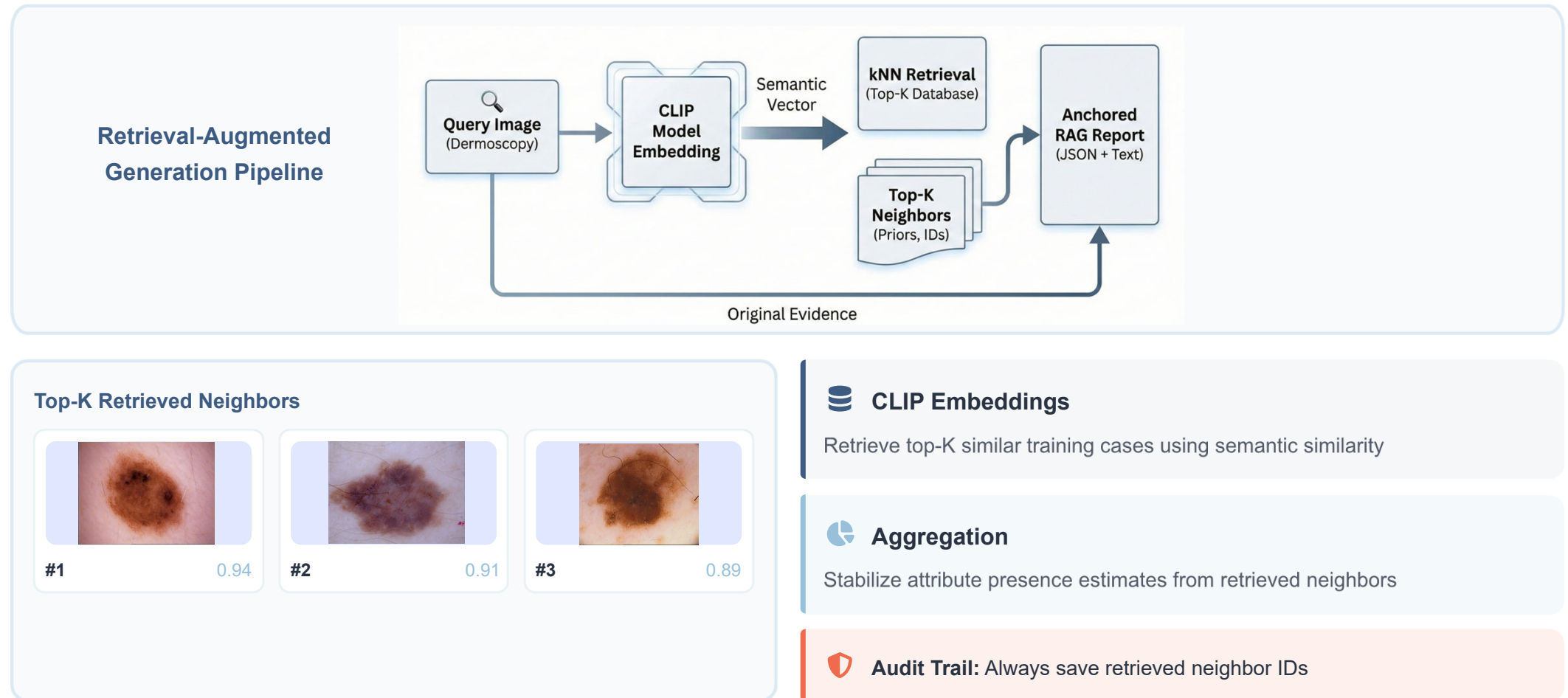
**Strict Schema**  
Controlled JSON structure

**Five Terms**  
All must be included

**Evidence-Based**  
Grounded in predictions

**Clear Communication**  
Clinical readability

## Bonus: CLIP semantic retrieval + RAG report





# Deliverables and evaluation

Each group only needs to submit **one** set of deliverables.

## Required Deliverables



### Lesion Masks

PNG files (one per image)



### Attribute Masks

PNG files in 5 separate folders



### Findings Report

One JSON per image and CSV with reports



### Bonus Audit File

CSV with top-K neighbor IDs, refined reports, and one JSON per image



### Technical Report

## Scoring Breakdown

### Segmentation Quality

25%

### Attribute Quality

25%

### Report Consistency

20%

### Bonus Retrieval Impact

10% (capped)

### Technical Report & Presentation

30%



**Important:** Ensure reproducible code and clear submission format

Code must also be submitted and may be randomly checked to verify reproducibility and compliance with the project rules.

# Recommended two-week plan and support

## 2-4 Setup & Baseline

Environment setup, data exploration, baseline segmentation model

## 4-7 Improve Segmentation

Model improvements, validation, qualitative checks

## 8-10 Attribute Detection

Train and evaluate five attribute masks

## 1-11 Report Generation

Anchored report generation and consistency validation

## 11 Bonus & Submission

CLIP retrieval integration, final submission, project report write-up, presentation



### Technical Tutorial

Separate hands-on session covering computer vision basics



### Q & A

Tutors will attend regular Q&A sessions



### Submission Checklist

Clear guidelines for deliverables.

 **Start with a strong baseline, iterate often, ask questions early!**

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# Technical report

Main sections:

- Literature review
- Methods
- Result and discussion
- Future work

No more than **12 pages** (exclude references)

Report template (not compulsory): <https://www.overleaf.com/latex/templates/imperial-college-london-msc-future-power-networks-unofficial-template/brrqsztycbyy>

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## Important Dates

- Submit the literature review on skin lesion segmentation: **27 July**
- Release of test set: **1 PM, 30 July**
- Submit the code, results, and final report: **7 PM, 30 July**
- Final presentation: **31 July**

Two awards will be given after the final presentation on 31 July:

- **Best Overall**
- **Best Segmentation Model**